INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

POWDERS, METAL (FOR USE IN AMMUNITION)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 4 types of atomized metal powders and one type of ellipsoidal metal powder. (see 6.3)

1.2 Classification. The metal powders are of the following types as specified (see 6.1 and 6.2):

Type I	-	Magnesium Powder, Atomized.
Type II	-	Aluminum Powder, Atomized.
Type III	-	Magnesium-Aluminum Alloy Powder 65/35, Atomized.
Type IV	-	Magnesium Powder, Ellipsoidal.
Type V	-	Aluminum Powder, Atomized for use in the rocket motor for the M913.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, Attn: RDAR-EIQ-SE, Picatinny Arsenal, New Jersey 07806-5000 or emailed to usarmy.picatinny.ardec.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1234 - Pyrotechnics: Sampling, Inspection and Testing
MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://quicksearch.dla.mil/ or from the Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E34 - Standard Test Method for Chemical Analysis of Aluminum and Aluminum Base Alloys
ASTM B92/B92M - Standard Specification for Unalloyed Magnesium Ingot and Stick for Remelting [Metric]
ASTM B215 - Standard Practices for Sampling Metal Powders
ASTM E300 - Standard Practice for Sampling Industrial Chemicals
ASTM B329 - Standard Test Method for Apparent Density of Metal Powders and Compounds Using the Scott Volumeter
ASTM B330 - Standard Test Methods for Estimating Average Particle Size of Metal Powders and Related Compounds Using Air Permeability

(Copies of ASTM standards are available online at http://www.astm.org or from American Society for Testing and Materials, 100 Barr Harbor Dr., West Conshohocken, PA 19428.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria)

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3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Material. Type I, III and IV magnesium powder shall be manufactured from magnesium ingot or stick that meets the requirements of ASTM B92M.

3.3 Shape.

3.3.1 Shape - Type I, III and V. The shape of the metal powders Type I, III and V particles shall be spheroidal (see 6.3).

3.3.2 Shape - Type II. The shape of the metal powders atomized Type II particles shall be spheroidal or granular (see 6.3).

3.3.3 Shape - Type IV. The shape of the ellipsoidal magnesium shall be spheroidal with a maximum of 30 percent oblong shape particles (see 6.3).

3.4 Chemical and physical properties. The metal powder shall conform to the chemical and physical requirements as shown in Table I.

3.4.1 Magnesium. The magnesium shall conform to the chemical and physical requirements as shown in Table I.

3.4.2 Aluminum. The aluminum shall conform to the chemical and physical requirements as shown in Table I.

3.4.3 Volatile material at 105° C. The volatile material at 105° C shall conform to the chemical and physical requirements as shown in Table I.

3.4.4 Oil and grease. The oil and grease shall conform to the chemical and physical requirements as shown in Table I.

3.4.5 Zinc. The zinc shall conform to the chemical and physical requirements as shown in Table I.

3.4.6 Silicon. The silicon shall conform to the chemical and physical requirements as shown in Table I.

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3.4.7 Total Fe. The total Fe shall conform to the chemical and physical requirements as shown in Table I.

3.4.8 Carbides. The carbides shall conform to the chemical and physical requirements as shown in Table I.

3.4.9 Total Al and Mg. The total aluminum and magnesium shall conform to the chemical and physical requirements as shown in Table I.

3.4.10 Apparent density. The apparent density shall conform to the chemical and physical requirements as shown in Table I.

3.5 Granulation. The metal powder shall conform to the granulation requirements in Table II.

3.6 Average particle size. The average particle size shall conform to the requirements in Table III.

3.6.1 Particle size for type V. The average particle size for type V aluminum powder shall be 25 ± 5 micrometers (µm).

3.7 Workmanship. The metal powder shall be uniform in quality and free from lumps, grit, visible impurities, foreign matter, or other defects that would render the metal powder unsuitable for intended use.

3.8 Process verification.

3.8.1 Type I, II,III and V. Type I, II, III and V powder shall be manufactured by the atomizing process.

3.8.2 Type IV. Type IV magnesium powder shall be manufactured by hammering of the material.

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TABLE I. Chemical and physical properties.

Requirement	I Magnesium Powder	II Aluminum Powder	III 65/35 Mg/Al Alloy Powder	IV Magnesium Ellipsoidal Powder	V Aluminum Powder
Magnesium, % min	98.0	---	65 ± 5	98.0	---
Aluminum, % min	---	98.75	35 ± 5	---	99.00
Volatile material at 105°C, % max	0.1	0.1	0.1	0.1	0.1
Oil and grease, % max	0.02	0.2	0.2	0.02	0.2
Zinc, % max	---	0.3	---	---	---
Silicon, % max	---	0.25	0.2	---	---
Total Fe, % max	0.22	0.55	0.1	0.22	0.20
Carbides, % max	0.004	---	---	0.004	---
Total Al and Mg, % max	---	---	98.0	---	---
Apparent density, g/mL (min)	0.95	1.0	1.0	0.95	---

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TABLE II. Granulation.

Type I Magnesium							
		Percent **Passing** By Weight					Percent **Retained** on
	Nominal mesh size	20/50	30/50	50/100	100/200	200/325	30/40
U.S. Standard Sieve No.	16	100 min	---	---	---	---	---
	20	95 min	95 min	---	---	---	0.0 - 0.1 max
	30	70 min	85 min	95 min	100 min	---	0.0 - 7.0 max
	40	---	---	---	---	---	43 - 53
	50	20 max	40 max	70 min	90 min	---	40 - 56
	60	10 max	25 max	---	---	---	0 - 3.0
	Thru 60	---	---	---	---	---	0 - 0.1
	80	---	---	---	---	---	---
	100	---	5 max	40 max	50 min	---	---
	140	---	---	20 max	---	---	---
	200	---	---	5 max	30 max	99.5 min	---
	325	---	---	---	5 max	60.0 min	---

Type II Aluminum							
		Percent **Passing** By Weight					
	Nominal mesh size	20/50	30/40	30/50	50/100	100/200	200/325
U.S. Standard Sieve No.	200	---	---	---	---	---	99.5 min
	325	---	---	---	---	---	60.0 min

Type III 65/35 Magnesium Aluminum Alloy							
		Percent **Passing** By Weight					
	Nominal mesh size	20/50	30/40	30/50	50/100	100/200	200/325
U.S. Standard Sieve No.	30	---	---	---	---	100 min	---
	50	---	---	---	---	90 min	---
	100	---	---	---	---	50 min	---
	200	---	---	---	---	30 max	99.5 min
	325	---	---	---	---	5 max	60.0 min

Type IV Ellipsoidal Magnesium							
		Percent **Retained** On					
	Nominal mesh size	20/50	30/40	30/50	50/100	100/200	200/325
U.S. Standard Sieve No.	20	0.0 - 0.1 max	---	0.1 max	---	---	---
	30	0.0 - 16.0 max	---	7.0 max	---	---	---
	40	45 - 65	---	43-53	---	---	---
	50	20 - 35	---	40-56	---	---	---
	60	0 - 3.0	---	0-3	---	---	---
	Thru 60	0 - 0.1	---	0 - 0.1	---	---	---

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TABLE III. Average particle size, in micrometers.

Nominal Mesh Size	20/50	30/50	50/100	100/200	200/325
Powder Type					
I	450 ± 50	350 ± 50	200 ± 25	125 ± 25	22 ± 8
II	---	---	---	---	22 ± 8
III	---	---	---	125 ± 25	22 ± 8
IV	---	350 ± 50	---	---	---

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4. VERIFICATION

TABLE IV. Requirement/verification cross reference matrix.

Method of verification		Classes of verification						
1 – Analysis 2 – Demonstration 3 – Examination 4 – Test		A – First Article B – Conformance						
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X	X	X	X	X		4.2
3.1.2	Conformance	X	X	X	X		X	4.3
3.2	Material	X		X	X	X	X	4.5.1
3.3.1	Shape - Type I, III and V			X		X	X	4.5.2
3.3.2	Shape - Type II			X		X	X	4.5.2
3.3.3	Shape - Type IV			X		X	X	4.5.2.1
3.4	Chemical properties				X	X	X	4.5.3 - 4.5.12, 4.5.14, 4.5.16
3.4.1	Magnesium				X	X	X	4.5.6 (Type I and IV), 4.5.16 (Type III)
3.4.2	Aluminum				X	X	X	4.5.10 (Type II, III and IV), 4.5.11 (Type III)
3.4.3	Volatile material at 105°C				X	X	X	4.5.3
3.4.4	Oil and grease				X	X	X	4.5.8
3.4.5	Zinc				X	X	X	4.5.7
3.4.6	Silicon				X	X	X	4.5.9
3.4.7	Total Fe				X	X	X	4.5.5
3.4.8	Carbides				X	X	X	4.5.4
3.4.9	Total Al and Mg				X	X	X	4.5.12
3.4.10	Apparent density				X	X	X	4.5.14
3.5	Granulation				X	X	X	4.5.15
3.6	Average particle size			X		X	X	4.5.13
3.6.1	Average particle size for Type V			X		X	X	4.5.13
3.7	Workmanship			X		X	X	4.5.17
3.8	Process verification		X			X	X	4.5.18

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4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2)

b. Conformance inspection (see paragraph 4.3)

4.2 First article inspection. When specified, a sample shall be subject to first article verification in accordance with Table V.

4.2.1 First article quantity. The first article sample shall be a minimum10 kilograms of metal powder from the first production lot produced by the contractor using the same production equipment, process procedures, and material supply sources as will be used in regular production.

4.2.2 Inspections to be performed. The first article sample shall be subjected to inspection and tests specified in Table V and 4.4.

4.2.2.1 Rejection. If any sample fails to comply with any of the applicable requirements the first article quantity shall be rejected.

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TABLE V. First article inspection.

Classification	Metal Powder	Requirement Paragraph	Inspection Method Reference
Critical	None defined		
Major			
101	Material	3.2	4.5.1
102	Shape	3.3	4.5.2, 4.5.2.1
103	Volatile material	3.4.3	4.5.3
104	Carbides	3.4.8	4.5.4
105	Total Fe	3.4.7	4.5.5
106	Magnesium	3.4.1	4.5.6 (Type I and IV), 4.5.16 (Type III)
107	Zinc	3.4.5	4.5.7
108	Oil and grease	3.4.4	4.5.8
109	Silicon	3.4.6	4.5.9
110	Free metallic aluminum	3.4.2	4.5.10 (Type II, III and V), 4.5.11 (Type III)
111	Total aluminum and magnesium for Type III powder	3.4.9	4.5.12
112	Average particle size	3.6	4.5.13
113	Apparent density	3.4.10	4.5.14
114	Granulation	3.5	4.5.15
Minor			
201	Workmanship	3.7	4.5.17
202	Process verification	3.8	4.5.18

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Lot formation shall be in accordance with the lot formation requirements of MIL-STD-1916. A lot shall consist of no more than one type of powder and no more than one granulation.

4.3.2 Classification of characteristics. The characteristics shown in 3.2 through 3.8 when tested in accordance with 4.5 shall constitute minimum inspections to be performed.

4.3.3 Inspection requirements by classification of characteristics.

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a. For the conformance inspection paragraph 4.3.3.1, the definitions of critical, major and minor defects are provided in paragraph "Definitions" of MIL-STD-1916.

4.3.3.1	Metal powder.	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical	None defined			
Major				
101	Material	4.4.1	3.2	4.5.1
102	Shape	4.4.1	3.3	4.5.2, 4.5.2.1
103	Volatile material	4.4.1	3.4.3	4.5.3
104	Carbides	4.4.1	3.4.8	4.5.4
105	Total Fe	4.4.1	3.4.7	4.5.5
106	Magnesium	4.4.1	3.4.1	4.5.6 (Type I and IV), 4.5.16 (Type III)
107	Zinc	4.4.1	3.4.5	4.5.7
108	Oil and grease	4.4.1	3.4.4	4.5.8
109	Silicon	4.4.1	3.4.6	4.5.9
110	Free metallic aluminum	4.4.1	3.4.2	4.5.10 (Type II, III and V), 4.5.11 (Type III)
111	Total aluminum and magnesium for Type III powder	4.4.1	3.4.9	4.5.12
112	Average particle size	4.4.2	3.6	4.5.13
113	Apparent density	4.4.1	3.4.10	4.5.14
114	Granulation	4.4.2	3.5	4.5.15
Minor				
201	Workmanship	4.4.2	3.7	4.5.17
202	Process verification	4.4.1	3.8	4.5.18

4.4 Metal powder sampling.

4.4.1 Sampling requirements. Two independent representative samples, minimum two kilograms, shall be obtained from each lot of magnesium powder using ASTM E300 or ASTM B215. Unless otherwise specified, sampling shall be conducted in accordance with Table VI. When lots comprise three or less containers, all containers in the lot shall be selected. Before sampling, the contents of each container (selected for sampling) shall be thoroughly blended for 10 minutes by rolling and turning end for end. Failure of any test, by any sample, shall be cause for rejection of the lot represented.

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TABLE VI. Sampling.

Containers in a Lot	Number of Samples
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
20,001 to 35,000	15
over 35,000	20

4.4.2 Sampling for granulation, particle size and workmanship. Each container shall be rolled and tumbled for ten minutes or until the metal powder is completely mixed in the container. Then a sample of minimum 227 grams shall be removed from the sample with a sampling thief. The granulation and particle size shall be determined on the sample thus obtained. Each container shall be individually tested. If any sample fails to comply with the requirements of Table II and Table III the container shall be rejected.

4.4.2.1 Alternate sampling for granulation, particle size and workmanship. The composite sample prepared in accordance with 4.4.1 may be used in lieu of the samples required in 4.4.2. If the composite sample fails to comply with the requirements of Table II and Table III the lot shall be rejected, or each container may be sampled in accordance with 4.4.2 for acceptance determination and if any container fails to comply with the requirements of Table II and Table III the container shall be rejected.

4.5 Test methods and procedures.

4.5.1 Material inspection. Material shall be inspected in accordance with the “Methods of Analysis” section of ASTM B92M.

4.5.2 Shape. Thoroughly mix the sample until it is homogeneous and withdraw a 10 gram sample for examination, then place the sample into a container, cover, and tumble the container for five minutes. With a scoop, remove a ¼ gram sample from the container and sprinkle the sample on a microscopic slide and examine under a 20 to 100 power microscope.

4.5.2.1 Shape for ellipsoidal type IV. Thoroughly mix the sample until it is homogeneous and withdraw a 10 gram sample for examination, then place the sample into a container, cover, and tumble the container for five minutes. With a scoop, remove a ¼ gram sample from the container and sprinkle the sample on a microscopic slide. The slide shall be placed on a stereo microscope (magnification may be 16 x to 24 x range) equipped with an eyepiece micrometer (photographic). Light shall be transmitted through the top and bottom of the sample. The length and width of 100 particles shall be measured. For each particle, the length shall be divided by the width to obtain the L/W oblong shape ratio. The percentage of particles with an L/W ratio of greater than 1.5 shall be determined.

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4.5.3 Volatile material at 105ºC. Approximately 5 grams of the sample shall be weighed into a tared dish and heated in an oven for 3 hours at 105ºC. Remove from oven and place into a dessicator to cool. Reweigh sample and the loss in weight shall be calculated as percent volatile material at 105ºC.

$$\text{Percent volatiles} = \frac{A \cdot 100}{B}$$

Where:

A = loss in weight, gm.
B = weight of sample, gm.

4.5.4 Carbides.

4.5.4.1 Gas chromatography with flame ionization detection (GC/FID).

4.5.4.1.1 Overview. The following procedure is used to determine the amount of magnesium carbide (acetylide) in magnesium. Sample is dissolved in 5 % sulfuric acid and acetylene thus produced is measured by GC/FID. The method can be extended to measure other types of carbides present provided that appropriate standards are available and the instrument method is modified appropriately.

4.5.4.1.2 Reagents.

a. Sulfuric acid (H_2SO_4), concentrated 95 – 98 % wt, ACS reagent grade.
b. Distilled water.
c. Sulfuric acid, 5 % wt aqueous solution, made by dissolving 30 milliliters of concentrated sulfuric acid in enough distilled water to make 1000 milliliters of solution. Scale the amounts proportionally to make more or less reagent depending on the number or samples to be run. Quantity: 100 milliliters per sample and 50 milliliters per standard.

4.5.4.1.3 Gas standard.

a. Compressed gas cylinder containing 1 mole % each C_1-C_2 hydrocarbons, i.e. methane, acetylene, ethylene, and ethane, e.g. Sigma 23462 or equivalent.
b. Matching regulator required; see supplies and equipment (4.5.4.1.4).

4.5.4.1.4 Supplies and equipment.

a. GC column: Agilent J&W CarbonPLOT 60 m/0.32 mm/1.5micometers (length/diameter/film thickness) or equivalent capable of resolving C_1-C_2 hydrocarbons in less than 10 minutes. Quantity: 1.
b. Round-bottom-3-neck flask, 500 milliliter, 24/40 ground-glass ports, e.g. Kontes 606000-1024 or eq. Quantity: 1.
c. 24/40 ground-glass barbed adapter, e.g. Kontes 182900-2440 or eq. Quantity: 2.
d. Addition funnel with stopcock, >50 milliliter, 24/40 ground glass joint, e.g. Ace Glass 7257-50 or eq. Quantity: 1.

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e. Valve, 3-way, glass stopcock, e.g. Kontes 843900-0002 or eq. Quantity: 1.

f. Rubber stopper/septum fitting barbed adapter, above, e.g. Fisher 20190 or eq. Quantity: 1 per sample run.

g. Gas regulator, single stage CGA-165 with syringe adapter, e.g. Sigma 501395 or eq. Quantity:

h. Powder funnel, preferably metal, with 5 centimeter long stem fitting 24/40 port or eq. made from aluminum foil. Quantity: 1.

i. Syringe, 0.1 milliliter, gas-tight with locking valve and 22 or 23 gauge, 2 inch needle, e.g. SGE 100R-V-GT or eq. Quantity: 1, spares recommended.

j. Syringe, 25 milliliter, gas-tight with locking valve and Luer lock connection, e.g. SGE 25MR-VLLMA-GT or eq. Quantity : 1.

k. Needle, 22 or 23 gauge, 2 inch length, fitting Luer lock connection, e.g. SGE 039802 or eq. Quantity: 1, spares recommended.

l. Silicone grease, high vacuum. Quantity: approximately 5 grams.

m. Support stand with clamp assortment. Quantity: 1, clamps as needed.

n. Tubing, thick-wall rubber, 1/4 inch ID. Quantity: approximately 1 foot.

o. Tubing, thick-wall rubber, 1/8 inch ID. Quantity: approximately 1 foot.

p. Union, barbed, reducing 1/4 in to 1/8 inch. Quantity: 1

q. Volumetric glassware, assorted. Quantity: as needed.

r. Gas chromatograph (GC) with flame ionization detector (FID) and computer/integrator. Compressed air, hydrogen, and helium supply as recommended by GC manufacturer. Quantity: 1

s. Vacuum pump or other vacuum source, 5 kiloPascal ultimate vacuum or better in less than 5 minutes, e.g. Sartorius 16615 or eq. Quantity: 1

t. Laboratory balance, ≥ 200 gram capacity, 0.001 gram resolution or better. Quantity: 1

u. Manometer, digital, range: 0 to ≥200 kiloPascal, resolution: 0.2 kiloPascal or better, e.g. Omega HHP4230 or eq. Quantity: 1

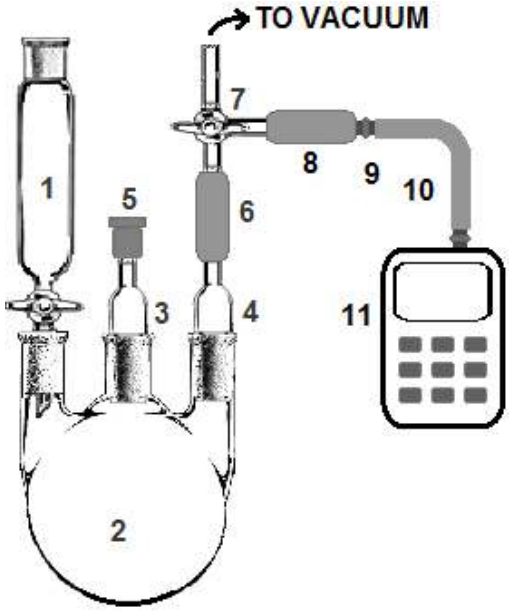
v. Room thermometer. Quantity: 1

w. Barometer, digital or analog, 0.2 kiloPascal resolution or better. Quantity: 1

4.5.4.1.5 Procedure. Set up the GC. Install and condition capillary column as required by manufacturer. The following analysis conditions are recommended: oven maintained at 140°C for a 5 minute analysis time, injector at 125°C flowing helium at 2 mL/min with a 20-fold split, FID at 250°C and maximum sensitivity setting and burning a mixture of hydrogen and air, fed at 40mL/min 400 mL/min, respectively. Assemble reaction apparatus as illustrated in Figure 1. Lubricate with silicone grease all moving parts and all joints and connections; remove excess grease. Support assembly with clamps and stand. Test gas-tightness of the assembled glassware. Evacuate the vessel and manometer connection using vacuum pump. Turn 3-way valve to exclude the vacuum pump and connect manometer and vessel. Observed pressure reading

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should be less than 10 kiloPascal and should not increase faster than 2 kPa/min. Check connections and eliminate leaks.



1. Addition funnel with stopcock, >50 mL, 24/40 ground glass joint,
2. Round-bottom-3-neck flask, 500 mL, 24/40 ground-glass ports,
3. 24/40 ground-glass barbed adapter,
4. 24/40 ground-glass barbed adapter,
5. Rubber stopper/septum,
6. Tubing, thick-wall rubber, 1/4 inch ID,
7. Valve, 3-way, glass stopcock,
8. Tubing, thick-wall rubber, 1/4 inch ID,
9. Union, barbed, reducing 1/4 in to 1/8 inch,
10. Tubing, thick-wall rubber, 1/8 inch ID,
11. Manometer, digital, 0 to ≥ 30 psi.

FIGURE 1. Reaction apparatus set-up & connections.

4.5.4.1.6 Blank.

a. Add 50.0 milliliters of 5 % wt sulfuric acid reagent to the addition funnel.

b. Evacuate reaction vessel, then turn 3-way valve to exclusively connect the flask and manometer.

c. Open addition funnel to allow acid to dispense; keep stopcock open long enough for air to enter flask and internal pressure to reach atmospheric.

d. Wait 5 minutes for the flask to warm to room temperature and open and close the stopcock to equilibrate pressure.

e. Puncture the rubber septum with the 0.1 milliliter syringe.

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f. With the syringe valve open and needle inserted through the septum, slowly pump the plunger 5 - 10 times to fill and empty the syringe.

g. Fill the syringe with 100 microliters of gas from the flask, close the syringe valve and remove the needle from the septum.

h. When ready to make injection into GC, depress the plunger of the syringe to 80 microliters and open the syringe valve.

i. Depress the plunger further to 50 microliter and quickly inject the syringe contents onto the GC and run the sample.

j. Once GC run is complete, take a second sample from the flask and inject the same way (4.5.4.1.6 e. through i.).

k. Identify and integrate the acetylene peak in each chromatogram. Under the recommended conditions, C_1-C_2 peaks should elute between 3 and 4 minutes in order of increasing molecular weight, however the blank may not show any signal above noise. Acetylene is the second fastest-eluting peak. Average the acetylene peak areas and label this value "A". Keep the sulfuric acid in the flask for preparation of calibration standard.

4.5.4.1.7 Calibration standard.

a. Attach regulator to reference gas cylinder. Check for leaks and proper operation.

b. Open main cylinder valve. Lower the outlet pressure by turning the regulator knob completely counter-clockwise. The gauge of a single stage regulator displays the gas pressure inside the cylinder, not the outlet pressure.

c. Attach needle to 25 milliliter syringe, open syringe valve and insert needle through regulator's syringe adapter.

d. Slowly turn the regulator knob clockwise until the plunger begins to move and syringe begins to fill. Fill and empty the syringe 5 times, then fill with 25 milliliters of reference gas, close syringe valve and withdraw the needle.

e. Evacuate flask headspace using the vacuum pump and turn 3-way valve to exclusively connect flask and manometer.

f. Record the atmospheric pressure and label it "B".

g. Depress 25 milliliter syringe plunger to 20 milliliters and open the valve, depress plunger further to a volume between 10 and 12 milliliters, note the exact volume "C" to nearest 0.2 milliliter, and insert needle through reaction flask septum and inject syringe contents. Then, withdraw the needle from the reaction flask septum.

h. Open addition funnel stopcock to let flask fill with air.

i. Close stopcock, allow 10 minutes for flask to warm to room temperature, then open and close the valve to equilibrate pressure.

j. Withdraw a 100 microliter sample from the flask and inject 50 microliters of it onto the GC in the same manner as the blank (4.5.4.1.6 e. through i.). Repeat twice more.

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k. Identify and integrate the acetylene peak in each chromatogram. Average the acetylene peak areas and record this value as “D”.

l. Disassemble reaction vessel and dispose of the sulfuric acid. Rinse the flask with distilled water, then with acetone and air dry completely.

4.5.4.1.8 Sample analysis.

a. Insert the two 24/40 barbed adapters into the flask, but not the addition funnel.

b. Pre-weigh powder funnel and weighing paper.

c. Weigh (to the nearest 0.001 gram) a 0.500 ± 0.020 gram aliquot of sample magnesium.

d. Transfer sample into the flask using the powder funnel, making sure no powder gets on the ground-glass joint.

e. Re-weigh the powder funnel and weighing paper to determine the weight “E” of sample transferred to the flask.

f. Completely assemble the reaction apparatus, and transfer 50.0 milliliters of 5 % wt sulfuric acid to the addition funnel.

g. Evacuate the flask using vacuum pump.

h. Turn 3-way valve to connect flask and manometer exclusively.

i. Open addition funnel stopcock to dispense acid into reaction flask; shut the stopcock immediately after all the acid has dispensed. Wait 15 minutes after bubbling completely stops in the reaction flask.

j. Record the flask internal pressure as “F”, and the atmospheric pressure “G”.

k. Withdraw two 100 microliter samples from the flask and inject 50 microliters of each onto the GC using the same technique as the blank (4.5.4.1.6 e. through i.).

l. Average acetylene peak areas and record the value as “H”.

m. Rinse the flask with distilled water, then with acetone and air dry completely.

n. Analyze a duplicate aliquot of magnesium the same way (4.5.4.1.8 a. through m.).

4.5.4.1.9 Sample calculation. Evaluate the expression below for each of the two aliquots and average the results to calculate carbide content in terms of magnesium carbide (acetylide). Report the average of the two aliquots.

$$\%\ \text{wt}\ MgC_2 = \frac{0.581 \cdot B \cdot C \cdot F \cdot y \cdot (H - A)}{E \cdot G \cdot T \cdot (D - A)}$$

Where:

A = Average acetylene peak areas. (see 4.5.4.1.6, k.).
B = Atmospheric pressure, (bar). (see 4.5.4.1.7, f.)
C = Exact volume, (mL). (see 4.5.4.1.7, g.)

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D = Average acetylene peak areas, standard. (see 4.5.4.1.7, k.)
E = Weight, (grams). (see 4.5.4.1.8, e.)
F = Flask internal pressure, (bar). (see 4.5.4.1.8, j.)
G = Atmospheric pressure, (bar). (see 4.5.4.1.8, j.)
H = Average acetylene peak areas. (see 4.5.4.1.8, l.)
T = Average ambient temperature, (Kelvin).
y = Mole percentage of acetylene in the reference gas. (see 4.5.4.1.9)

Besides magnesium acetylide, the expression can be used to find the amount of carbides, in terms of magnesium acetylide equivalent, that yield methane, ethene, ethane and other light hydrocarbons; substitute the appropriate "y" for each type and multiply the end result by the ratio {number of carbons in the hydrocarbon to that in MgC_2 (two)}.

4.5.4.2 Carbides (alternate 1).

4.5.4.2.1 Apparatus. Apparatus similar to that shown in Figure 2 shall be used. There must be a source of inert gas such as nitrogen, argon, or carbon dioxide.

4.5.4.2.2 Reagent. Hydroxylamine hydrochloride reagent. Dissolve 0.25 grams of gelatin in hot distilled water. Dilute to 500 milliliters and add 500 milliliters of 95 % ethanol (or isopropanol) and 1.25 grams of hydroxylamine hydrochloride.

4.5.4.2.3 Procedure.

a. Add an accurately weighed portion of approximately 50 grams of magnesium powder to the reaction flask (A).
b. Transfer 50 milligrams of cuprous chloride to the test tube (B).
c. Pass nitrogen (or other inert gas) through the apparatus to remove the air and continue the passage of nitrogen throughout the determination.
d. Disconnect the test tube and add into it 15 milliliters of concentrated NH_4OH and 30 milliliters of the hydroxylamine hydrochloride reagent.
e. Reconnect the test tube.
f. Add 100 milliliters of distilled water from dropping funnel (C) and switch on the heater.
g. Heat the mixture in the reaction flask just to boiling.
h. A red or pink coloration of the cuprous ammonium chloride solution in the test tube indicates that the magnesium sample contains more than 0.004 percent carbides. Should the cuprous ammonium solution become blue due to the introduction of air, it must be decolorized with successive 5 milliliter portions of hydroxylamine hydrochloride reagent until the blue color disappears.

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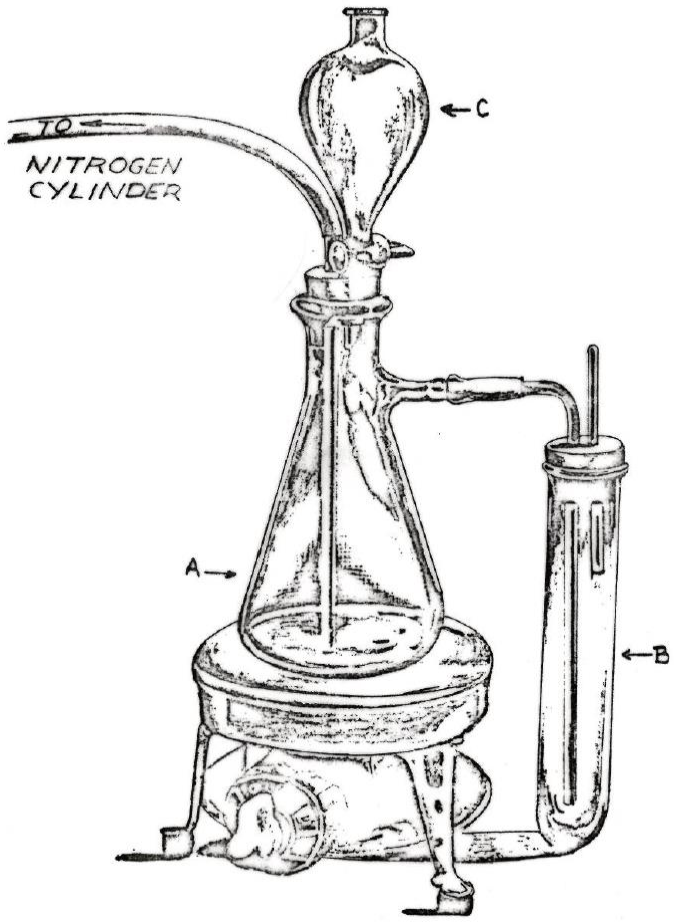


FIGURE 2. Apparatus for determination of carbides in magnesium.

4.5.5 Total Fe (ICP method).

4.5.5.1 Reagents. Ultra high purity acids shall be used for sample dissolution. It is advisable to use commercially available stock solutions with certified concentrations traceable to primary standards (National Institute of Standards and Technology or international measurement standards). Stock solutions shall be replaced after one year or in accordance with manufacturer recommendations. Alternately, stock solutions may be prepared in-house from high purity metals or their salts. The maximum recommended shelf life is one year from date of initial preparation.

a. Certified Fe stock solution shall contain 100.00 µg/mL Fe in 5.00 % 1:1 HNO_3 (v/v).
b. Certified Mg stock solution shall contain 20,000.00 µg/mL Mg in 0.10 % 1:1 HNO_3 (v/v).
c. Nitric Acid (1:1 HNO_3), concentrated, 67 – 70 %.
d. Distilled water.

4.5.5.2 Procedure.

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4.5.5.2.1 Sample Dissolution.

a. Weigh one gram of sample, as "M", to the nearest 1 milligram and transfer into 400 milliliter beaker.
b. Add 10 milliliters of distilled water to the beaker followed by 2 milliliters of 1:1 HNO_3. (Slowly add acid and allow reaction to subside.)
c. Gradually add 16 milliliters of 1:1 HNO_3 to the sample solution, approximately 1 milliliter at a time. Reaction may become overly vigorous by adding reagent too quickly.
d. Cover with watch glass and heat on hot plate at approximately 180°C to completely decompose sample.
e. Allow sample to cool to ambient temperature and transfer into 100 milliliter volumetric flask.
f. Rinse lower surface of watch glass and inner walls of beaker with distilled water into flask.
g. Dilute the sample solution with distilled water to 100 milliliters.

4.5.5.2.2 Preparation of calibration curve.

a. Dispense 50 milliliters of Mg stock solution into each of two 100 milliliter volumetric flasks.
b. Add stepwise 20.0 milliliters, and 30.0 milliliters of Fe stock solution separately to the two volumetric flasks and dilute with distilled water to 100 milliliters. These solutions will represent the mid and high standards respectively.
c. Prepare blank by pipetting 50 milliliters of Mg stock solution into a third 100 milliliter volumetric flask and dilute with distilled water to the 100 milliliter mark.

Mid standard shall contain: 20 µg/mL Fe and 10,000 µg/mL Mg.
High standard shall contain: 30 µg/mL Fe and 10,000 µg/mL Mg.
Blank shall contain 10,000 µg/mL Mg.

d. Aspirate standard solutions into ICP spectrometer in order of increasing concentration. Take emission measurements for iron and generate a calibration curve utilizing the instrument's software. Fe emission intensity shall be measured using radial view at the 259.940 nm wavelength. The calibration shall be forced through the origin. Repeat the calibration if the correlation determination (R_2) for Fe is < 0.9999.

4.5.5.2.3 Sample Analysis.

a. Aspirate sample solution into ICP spectrometer and record Fe emission measurement. Software generated concentrations for Fe shall be inserted into the formula below as A_1.
b. Aspirate blank solution containing 10,000 µg/mL Mg and insert into formula below as A_2.

4.5.5.2.4 Sample Calculation.

$$X = \frac{100 \cdot (A_1 - A_2)}{M}$$

Where:

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X = content of iron in sample [% (mass fraction)]
A_1 = amount of iron detected in the sample (g)
A_2 = amount of iron detected in the blank test solution (g)
M = amount of sample weighed out (g)

4.5.6 Magnesiun.

4.5.6.1 Magnesium (preferred method).

4.5.6.1.1 Reagents.

a. Chromic acid solution (approximately 0.1 N as acid). Dissolve 10.00 grams of CrO_3 in distilled water and dilute to 1000 milliliters.
b. Magnesium standard solution – Dissolve 1.8 grams of pure magnesium in 125 milliliters of 1:4 hydrochloric acid. Dilute to 1000 milliliters in a volumetric flask with distilled water. (One milliliter contains approximately 1.8 mg of magnesium). W_A = Actual weight of magnesium used in grams.
c. EDTA reagent solution – Dissolve 40.0 grams of disodium dihydrogen ethylenediaminetetracacetate dihydrate in distilled water and dilute with distilled water to 1000 milliliters in a volumetric flask.
d. Buffer solution (approximately pH 10) – Dissolve 65.5 grams of reagent grade ammonium chloride in 300 milliliters of distilled water. Add 570 milliliters of concentrated (28-30 % w/w) ammonium hydroxide and dilute to 1000 milliliters with distilled water.
e. Eriochrome Black T indicator solution – Dissolve 0.8 grams of Eriochrome Black T in a mixture of 40 milliliters methanol and 60 milliliters of triethanolamine.

4.5.6.1.2 Standardization.

a. Pipette 50.00 milliliters of magnesium standard into a 600 milliliter flask. Dilute to about 300 milliliters with distilled water.
b. Add 20 milliliters of buffer solution and 6 -10 drops of Eriochrome Black T indicator. Titrate with EDTA to a color change of red to blue against a reflected light background. There should be no red color remaining. V_A = Volume of EDTA used in milliliters.

$$X = \text{Milligrams of magnesium per milliliter EDTA} = 50 \cdot \frac{W_A}{V_A}$$

Where:

W_A = Actual weight of magnesium used in grams.
V_A = Volume of EDTA used in milliliters.

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4.5.6.1.3 Procedure.

a. Weigh approximately 1 gram (± 0.2 mg) of sample into a 250 milliliter beaker. W_B = weight of actual sample used.
b. Add 50 milliliters of 0.1 N chromic acid to the sample and allow to stand for 15 minutes, swirling the beaker every five minutes.Transfer the sample to a 30 milliliter, fine porosity sintered glass crucible and wash the sample with 25 milliliter portions of distilled water 6 times. The last washing should be colorless.
c. Wash the sample back into the original beaker with distilled water.
d. Treat the crucible with two 10 milliliter portions of 10 % hydrochloric acid to dissolve any metal remaining in it, returning the washings into the beaker with distilled water. Cover the beaker with a watch glass.
e. Add 10 milliliters of concentrated hydrochloric acid in small amounts to the sample. Gradually heat the beaker to 100°C maximum to dissolve any remaining metal.
f. Cool and transfer the solution to a 500 milliliter volumetric flask.
g. Dilute the solution to the 500 milliliter mark with distilled water and mix.
h. Pipette 50.00 milliliters of this solution into a 600 milliliter beaker.
i. Dilute the sample solution to 300 milliliters with distilled water.
j. Add 20 milliliters of buffer solution and 6 -10 drops of Eriochrome Black T indicator to the sample solution. Titrate with EDTA to a color change of red to blue against a reflected light background. There should be no red color remaining. V_B = Volume of EDTA used in milliliters.

$$\% \text{ metallic magnesium} = V_B \cdot \frac{X}{W_B}$$

Where:

X = Milligrams of magnesium per milliliter EDTA.
V_B = Volume of EDTA used in milliliters.
W_B = Weight of actual sample used.

4.5.6.2 Free metallic magnesium (eudiometer method) (alternate method). The free metallic magnesium shall be determined in accordance to method 412.1 of Standard MIL-STD-1234.

4.5.7 Zinc. Ammonium Mercuric Thiocyonate Method or Zinc Oxide Method. The zinc content shall be determined in accordance with ASTM E34. (see 6.6).

4.5.8 Oil and grease.

a. A weighed portion of approximately 20 grams of the sample shall be extracted with diethyl ether or hexane in a Soxhlet or similar extractor for minimum 6 hours using a tared flask.
b. The ether shall be evaporated by steam bath when extraction is completed in the flask.

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c. The contents shall be dried at 90°C, to constant weight, cooled in a desiccator, and weighed. A blank shall be run at the same time.
d. The weight of the residue shall be calculated to percent oil and grease, as follows:

$$\text{Percent oil and grease} = \frac{100 \cdot (A-B)}{W}$$

Where:

A = weight of ether extract.
B = weight of blank.
W = weight of sample.

4.5.9 Silicon as Si. Verify the silicon content by photometric examination (see 6.7).

4.5.9.1 Reagents.

a. Aluminum nitrate solution ($Al(NO_3)_3$) – Transfer 1.0 g of aluminum (99.9% purity or higher), low-silicon, to a 250-mL nickel beaker. Add 100 mL of NaOH solution and cover. Allow to react without applying heat until the reaction subsides, and then warm gently to complete the dissolution. Cool, and transfer to a 400 mL beaker containing 125 mL of HNO_3 (1+1). Cover and warm gently until the salts dissolve and the solution becomes clear. Cool, transfer to a 250 mL volumetric flask, dilute with distilled water to volume, and mix by inverting flask five (5) or more times.
b. Ammonium molybdate solution (100 g/L) – Dissolve 100 g of ammonium molybdate tetrahydrate ($(NH_4)_6Mo_7O_{24}$•4H2O) in water and dilute to 1 L. Store in a plastic bottle. Before using, filter through a plastic funnel and Whatman 41 filter paper or equivalent. If a precipitate forms on standing, the solution should be discarded.
c. Silicon, standard solution (1 mL = 0.05 mg Si) – Fuse 0.1070 g of anhydrous silicon dioxide (SiO_2) with 1.0 g of sodium carbonate (Na_2CO_3) in a covered platinum crucible. Cool, dissolve the melt completely in water in a plastic beaker, cover, and heat on a steam bath for ½ to 1 hour. Cool, transfer to a 1 L volumetric flask, dilute to volume, and mix by inverting flask five (5) or more times. Store the solution in a plastic bottle. If anhydrous SiO_2 is not available, the weight should be adjusted according to the actual silicon content of the SiO_2 used, as determined by gravimetric analysis.
d. Sodium hydroxide solution (NaOH) (300 g/L) – Dissolve 300 g of sodium hydroxide (NaOH) in about 600 mL of water, using a nickel or stainless steel beaker. Cool and dilute to 1 L. Store in a plastic bottle.
e. Nitric acid solution (HNO_3 (1+1)) – Thoroughly mix one part reagent-grade hydrogen nitrate (HNO_3(aqueous) (approximately 70%)) with one part deionized water.

4.5.9.2 Procedure.

4.5.9.2.1 Preparation of calibration curve.

a. Calibration solutions – Transfer 12.5 mL of $Al(NO_3)_3$ (see 4.5.9.1.a) solution and 1.0 mL of HNO_3 (1+1) to seven 100 mL beakers. Transfer 1.0, 2.0, 4.0, 8.0, 12.0, 16.0, and 20.0

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mL of silicon solution (1 mL = 0.05 mg Si) to each of the 100 mL beakers. Dilute to 80 mL and mix. Proceed as directed in 4.5.9.2.1.c.

b. Reference solution – Transfer 12.5 mL of $Al(NO_3)_3$ solution and 1 mL of HNO_3 (1+1) to a 100 mL beaker, dilute to 80 mL and mix. Proceed as directed in 4.5.9.2.1.c.
c. Color development – Using a pH meter, check the pH of the solution. If the pH is not within 1.1 to 1.3, adjust it to this range using HNO_3 (1+1) or NaOH solution as required. Add 10 mL of ammonium molybdate solution, transfer the solution to a 100 mL volumetric flask, dilute to volume, and mix. Let stand at least 5 minutes but no longer than 20 minutes before taking the photometric reading.
d. Photometry – Transfer a suitable portion of the reference solution to an absorption cell with a 2-cm light path and adjust the photometer to the initial setting, using a light band centered at approximately 400+. While maintaining this adjustment, take the photometric readings of the calibration solutions.
e. Calibration curve – Plot the photometric readings of the calibration solutions against milligrams of silicon per 100 mL of solution.

4.5.9.2.2 Sample analysis.

a. Test solution
   i. Transfer 0.10 g of the sample, weighted to the nearest 0.1 mg, to a 50 mL nickel crucible, add 10.00 mL of NaOH solution, and cover. Allow to react without applying heat until the reaction subsides, wash down the cover and sides of the crucible with a minimum amount of water, and boil gently until the reaction is complete and the silicon completely oxidized (see 6.7.1). Cool, dilute to approximately 30 mL with water, and transfer quantitatively to a 150 mL beaker containing 12.5 mL of HNO_3 (1+1). Cover and warm gently until the solution becomes clear (see 6.7.2-3). Cool, transfer to a 100 mL volumetric flask, dilute to volume, and mix by inverting flask five (5) or more times until solution is visually homogenous.
   ii. Transfer a 50 mL aliquot to a 100 mL beaker, add 1.0 mL of HNO_3 (1+1), dilute to approximately 80 mL, and mix by inverting flask five (5) or more times until solution is visually homogenous.
b. Reagent blank
   i. Transfer 10.0 mL of NaOH solution to a 100 mL volumetric flask containing 11.5 mL of HNO_3 (1+1). Dilute to approximately 50 mL, cool, dilute to volume, and mix. Transfer a 50 mL aliquot to a 100 mL beaker, add 1 mL of HNO_3 (1+1), dilute to 80 mL, and mix by inverting flask five (5) or more times until solution is visually homogenous.
   ii. To the remaining portion of the reagent blank add 1 mL of HNO_3 (1+1), dilute to volume, and mix. This portion is used as the reagent blank for the reference solution in 4.5.9.2.2.d.
c. Color development – Proceed as directed in 4.5.9.2.1.c.
d. Reference solution – To the remaining portion of the test solution add 1 ml of HNO_3 (1+1), dilute to volume, and mix. Take the photometric reading of the reference solution as directed in 4.5.9.2.1.d using the reagent blank obtained in 4.5.9.2.2.b.ii for the initial setting of the photometer.

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4.5.9.2.3 Calculation. Convert the photometric readings of the test and reference solutions to milligrams of silicon by means of the calibration curve. Calculate the percentage of silicon as follows:

$$Silicon, \% = \frac{(A-B)}{(C \times 10)}$$

Where:

A = silicon found in 100 mL of the final solution, mg.
B = reference solution correction, expressed as milligrams of silicon.
C = sample represented in 100 mL of the final solution, g.

4.5.10 Free metallic aluminum for Type II, Type III and Type V (eudiometer method). The free metallic aluminum for Type II, Type III and Type V shall be determined as specified in method 412.1 of Standard MIL-STD-1234. Except that the reaction flask shall contain 100 milliliters of 10 percent sodium hydroxide solution saturated with hydrogen. At standard conditions a 0.33 to 0.34 grams sample shall be used and calculated as follows:

$$\text{Percent free metallic aluminum} = \frac{0.0288 \cdot V \cdot (P_1 - P_2)}{W \cdot (273 + T)} - (0.275 \cdot A + 1.28 \cdot B)$$

Where:

T = temperature of water jacket in °C.
V = volume of gas in buret, milliliter.
P_1 = barometric pressure, in millimeter of mercury.
P_2 = vapor pressure of water at T, in millimeter, of mercury.
W = weight of sample, grams.
A = percent zinc.
B = percent silicon.

4.5.11 Aluminum, Test applicable to Type III, magnesium-aluminum alloy powdered 65/35 atomized.

4.5.11.1 Preparation of solutions.

4.5.11.1.1 8-Hydroxyquinoline solution. Five grams of 8-Hydroxyquinoline shall be dissolved in 10 milliliters of glacial acetic acid. The solution shall be diluted to 100 milliliters, with distilled water and filtered if necessary.

4.5.11.1.2 Bromphenol blue indicator. Into 20 milliliters of 95 percent ethyl alcohol 0.100 grams of bromphenol blue powder shall be dissolved and diluted to 100 milliliters with distilled water.

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4.5.11.2 Procedure.

a. An accurately weighed sample of 0.7700 plus or minus 0.0300 grams of Type III alloy shall be transferred to a 250 milliliter beaker.
b. Ten milliliters of distilled water shall be added to the sample beaker.
c. The beaker shall be covered with a watch glass and 20 milliliters of concentrated HCl shall be added dropwise through the opening at the lip of the beaker.
d. After the reaction in the beaker has subsided, the solution shall be heated to boiling to assure complete solution.
e. The watch glass shall be washed thoroughly with distilled water and the washings drained back into the beaker.
f. The solution shall be filtered through a No. 41 Whatman, or equivalent, filter paper with the filtrate caught in an accurately calibrated, 1 liter volumetric flask.
g. The filter paper shall be washed thoroughly with hot distilled water and the solution made up to the 1 liter volume with the distilled water.
h. A 100 milliliter aliquot shall be transferred to a 250 milliliter beaker using an accurately calibrated buret or pipet.
i. Three drops of bromphenol blue indicator shall be added to the solution neutralized by adding dropwise and with constant stirring, a filtered 1:1 NH_4OH solution until the yellow color turns to blue.
j. Two drops of NH_4OH shall then be added to the excess solution.
k. The solution shall be heated to boiling and 18 milliliters of the 8-hydroxyquinoline solution added in small portions, stirring after each addition to dissolve the precipitate that forms. Any precipitate remaining after the completion of the addition of the reagent shall be disregarded.
l. The solution shall be heated to boiling and 40 milliliters of 2 molar (M) ammonium acetate solution (154 grams per liter) added dropwise with vigorous stirring and then placed in a steam bath for 30 minutes.
m. The solution shall be filtered with a suction through a tared, fine porosity, fritted glass bottomed crucible.
n. The precipitate shall be washed eight times with 15 milliliter portion of distilled water, releasing the suction before each addition and allowing the precipitate to remain in contact with the wash water for at least one minute prior to the application of suction.
o. The precipitate shall be dried for 3 hours in an oven at 120° to 140°C, cooled in a desiccator and weighed as aluminum hydroxyquinoline, $Al(C_9H_6ON)_3$.

The percent aluminum shall be calculated as follows:

$$\text{Percent aluminum} = \frac{5.87 \cdot (A-(0.0874 \cdot B \cdot W))}{W}$$

Where:

A = grams of precipitate.
B = percent iron.
W = grams of sample contained in the aliquot.

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4.5.12 Total aluminum and magnesium. The percentage of aluminum and magnesium as determined in 4.5.11.1 and 4.5.16.1 shall be added to obtain the total aluminum and magnesium.

4.5.13 Determination of average particle size (all types). Determine the average particle size of the magnesium powder in accordance with the method described in ASTM B330 (see 6.8).

4.5.14 Apparent density. Determine the apparent density of the magnesium powder, as specified in table I, in accordance with the method described in ASTM B329.

4.5.15 Granulation (all types). The granulation shall be determined in accordance with Method 201.1 of Standard MIL-STD-1234 except using a mechanical shaker geared to produce 268 to 315 gyrations (see 6.8).

4.5.16 Test for magnesium percentage of Type III Magnesium/Aluminum Alloy powder 65/35, atomized.

4.5.16.1 Magnesium (preferred method).

a. A 100 milliliter aliquot of the solution, prepared as specified in 4.5.11.2 shall be transferred to a 400 milliliter beaker by means of an accurately calibrated buret or pipet.
b. Three drops of bromphenol blue indicator shall be added and the solution neutralized with filtered 1:1 NH_4OH solution until the yellow color turns to blue.
c. One gram of ammonium chloride and 1 milliliter of glacial acetic acid shall be added to the solution.
d. Twenty milliliters of hot 10 percent ammonium benzoate solution shall be added slowly to the solution while stirring.
e. The solution shall be heated on a hot plate and boiled gently for 5 minutes.
f. Filter the solution through a No. 41 Whatman, or equivalent, filter paper catching the filtrate in a 600 milliliter beaker.
g. The precipitate shall be washed 10 times with hot wash solution containing 1 gram of ammonium benzoate and 2 milliliter of glacial acetic acid per 100 milliliter of solution.
h. Allow the solution to cool to room temperature. Any crystallization that occurs in the filtrate as it cools shall be disregarded. The combined filtrate and washing shall be made alkaline to phenolphthalein with filtered concentrated NH_4OH solution.
i. Five milliliter of NH_4OH solution shall be added to the excess.
j. The solution shall be heated from 60° to 70°C and 15 milliliters of 8-hydroxyquinoline solution shall be added rapidly by drops from a buret. The liquid shall be stirred vigorously with a thermometer keeping the temperature of the solution at 60° to 70°C during addition.
k. The solution shall be heated almost to boiling with frequent stirring and transferred to a steam bath for 20 minutes.
l. The solution shall be filtered through a tared, fine porosity, sintered glass bottom crucible.
m. The suction shall be disconnected and 10 milliliters of warm 1:10 NH_4OH wash solution added to the precipitate with a wash bottle agitating the precipitate as much as possible with the force of the steam of the wash solution. The suction shall be applied until the

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precipitate remains only moist enough to prevent cracking. This procedure shall be repeated 7 times finally washing with two 10 milliliter portions of cold distilled water.

n. The precipitate shall be dried to constant weight in an oven at 155° to 160°C, cooled in a dessicator, and weighed.

The percent magnesium shall be calculated as follows:

$$\text{Percent magnesium} = \frac{7.78 \cdot A}{W}$$

Where:

A = grams of precipitate.
W = grams of sample contained in the aliquot.

4.5.16.2 Magnesium (alternate method).

4.5.16.2.1 Reagents.

a. Chromic acid solution (approximately 0.1 N as acid). Dissolve 10.00 grams of CrO_3 in distilled water and dilute to 1000 milliliters (mL).
b. Magnesium standard solution – Dissolve 1.8 grams of pure magnesium in 125 milliliters of 1:4 hydrochloric acid. Dilute to 1000 milliliters in a volumetric flask with distilled water. (One milliliter contains approximately 1.8 milligrams of magnesium). W_A = Actual weight of magnesium used in grams.
c. EDTA reagent solution – Dissolve 40.0 grams of disodium dihydrogen ethylenediaminetetracacetate dihydrate in distilled water and dilute with distilled water to 1000 milliliters in a volumetric flask.
d. Buffer solution (approximately pH 10) – Dissolve 65.5 grams of reagent grade ammonium chloride in 300 milliliters of distilled water. Add 570 milliliters of concentrated (28-30 % w/w) ammonium hydroxide and dilute to 1000 milliliters with distilled water.
e. Eriochrome Black T indicator solution – Dissolve 0.8 grams of Eriochrome Black T in a mixture of 40 milliliters methanol and 60 milliliters of triethanolamine.

4.5.16.2.2 Standardization.

a. Pipette 50.00 milliliters of magnesium standard into a 600 milliliter flask. Dilute to about 300 milliliters with distilled water.
b. Add 20 milliliters of buffer solution and 6 -10 drops of Eriochrome Black T indicator. Titrate with EDTA to a color change of red to blue against a reflected light background. There should be no red color remaining. V_A = Volume of EDTA used in milliliters.

$$X = \text{Milligrams of magnesium per milliliter EDTA} = 50 \cdot \frac{W_A}{V_A}$$

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Where:

W_A = Actual weight of magnesium used in grams.
V_A = Volume of EDTA used in milliliters.

4.5.16.2.3 Procedure.

a. Weigh approximately 1 gram (± 0.2 mg) of sample into a 250 milliliter beaker. W_B = weight of actual sample used.
b. Add 50 milliliters of 0.1 N chromic acid to the sample and allow to stand for 15 minutes, swirling the beaker every five minutes. Transfer the sample to a 30 milliliter, fine porosity sintered glass crucible and wash the sample with 25 milliliter portions of distilled water 6 times. The last washing should be colorless.
c. Wash the sample back into the original beaker with distilled water.
d. Treat the crucible with two 10 milliliter portions of 10 % hydrochloric acid to dissolve any metal remaining in it, returning the washings into the beaker with distilled water. Cover the beaker with a watch glass.
e. Add 10 milliliters of concentrated hydrochloric acid in small amounts to the sample. Gradually heat the beaker to 100°C maximum to dissolve any remaining metal.
f. Cool and transfer the solution to a 500 milliliter volumetric flask.
g. Dilute the solution to the 500 milliliter mark with distilled water and mix.
h. Pipette 50.00 milliliters of this solution into a 600 milliliter beaker.
i. Dilute the sample solution to about 300 milliliters with distilled water.
j. Add 20 milliliters of buffer solution and 6 -10 drops of Eriochrome Black T indicator to the sample solution. Titrate with EDTA to a color change of red to blue against a reflected light background. There should be no red color remaining. V_B = Volume of EDTA used in milliliters.

$$\% \text{ metallic magnesium} = V_B \cdot \frac{X}{W_B}$$

Where:

X = Milligrams of magnesium per milliliter EDTA.
V_B = Volume of EDTA used in milliliters.
W_B = weight of actual sample used.

4.5.17 Workmanship. The powder shall be visually examined with unaided eyes by personnel with normal vision or normal corrected vision. The powder shall be uniform in quality and free from lumps, grit, visible impurities, foreign matter or other defects that would render the powder unsuitable for the intended use.

4.5.18 Process verification.

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4.5.18.1 Type I, II, III and V. The manufacturing process shall be demonstrated to be manufactured use the atomizing process.

4.5.18.2 Type IV. The manufacturing process where the material is mechanically ground (or grinded) in a hammermill or other similar machine.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Magnesium powder covered by this specification is intended for use in the manufacture of tracer, igniter, signal flare and pyrotechnic compositions.

6.1.1 Type V aluminum powder. The type V aluminum powder outlined in this specification is intended for use in the rocket motor of the 105mm M913.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Type and metal powder (see 1.2).

c. Granulation and average particle size required (see 3.4 and 3.5)

d. Whether a first article sample is required (see 3.1.1).

e. Packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Atomization. Atomization is defined as the dispersion of a molten metal into a spray, the spheroidal droplets of which are then allowed to freeze under the primary influence of surface tension.

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6.3.2 Granular. Granular is defined as particles that have approximately equi-dimensional nonspherical shapes.

6.3.3 Ellipsoidal particles. Ellipsoidal particles are made by chipping magnesium ingots and then hammering the particles in a hammermill machine.

6.3.4 Oblong particles. An oblong particle is one that the length/width ratio is greater than 1.5.

6.4 Change history.

Specification Revision	ERR Number	ERR Release Date
MIL-P-14067	-	16 May 1955
MIL-P-14067 A	-	13 February 1963
MIL-P-14067 B	-	10 March 1967
MIL-P-14067 B Amend 3	-	15 November 1971
MIL-P-14067 B Amend 4	-	8 February 1999
MIL-DTL-14067 C	R13Q2081	9 May 2014
MIL-DTL-14067 C Amend 1	R15Q2009	---------

6.5 Units.

µm = micrometer
µg = microgram
g = gram
mg = milligram
mL = milliliter
min = minute
psi = pounds per square inch
kPa = kilopascal
m = meter
v/v = volume over volume
w/w = weight over weight
min = minimum
max = maximum

6.6 Alternate zinc test. ASTM E34, test methods 210 through 220, might be considered as alternate test procedures.

6.7 Silicon test method. The following notes are included for informational purposes regarding the silicon test method in 4.5.9. This test method originally appeared in ASTM E34.

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6.7.1 Achieving complete dissolution. With certain alloys it may be necessary to evaporate the caustic solution to pastiness in order to completely dissolve the silicon. Care should be taken to avoid mechanical loss by spattering.

6.7.2 Solution cloudiness and excessive heating. Cloudiness (gray) of the solution at this point usually indicates incomplete dissolution of the silicon and another portion of the sample must be taken. Prolonging the heating after the NaOH attack will probably correct the difficulty. However, excessively prolonged heating will tend to cause precipitation of the manganese.

6.7.3 Brownish-yellow solutions. If the sample contains high manganese, the solution may be colored brownish-yellow due to suspended manganese dioxide (MnO_2). To bleach the color, add a saturated sodium sulfite (Na_2SO_3) solution drop-wise to the hot solution. Avoid any appreciable excess. Add potassium permanganate ($KMnO_4$) solution (3.2 g/L) until the solution is tinted pink. Finally add just enough oxalic acid (10 g/L) to destroy the pink color. Transfer the solution to a 100-mL volumetric flask, dilute to volume, and mix.

6.7.4 Correction for reference solution. For routine work, the reference solution correction may be omitted unless the test solution is colored yellow.

6.7.5 Alternate silicon test. JIS H 1352, “ICP atomic emission spectrometry (method B)”, might be considered as an alternate test procedure.

6.8 Additional particle size requirements and analysis. Programs and suppliers may consider it desirable to impose additional requirements on particle size above and beyond those within the tolerances specified so as to specify further refinement of requirements or receive product optimized for their manufacturing processes. Furthermore, they may consider using alternate or additional methods and equipment to specify, characterize, and analyze size. Such analysis techniques may include:

a. Gravitational sedimentation (see ASTM B761 and ISO 10076)
b. Microscopy (see ISO 13322 and BS 3406-4)
c. Laser diffraction (see ASTM B882 and ISO 13320)

Those who are interested in alternate characterization techniques are cautioned that analysis and results from one technique are generally not directly comparable to other techniques due to numerous factors including particle morphology, measurement technique, etc. Beneficial information may be found on alternate techniques from the National Institute of Standards and Technology (NIST) Special Publication 960-1, Recommended Practice Guide on Particle Size Characterization, available at http://www.nist.gov.

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6.9 Subject term (key word) listing.

Magnesium
Aluminum
Granulation
Atomization
Ellipsoidal particles
Oblong particles

Custodians:
Army- AR
Navy- OS
Air Force- 11

Review Activities:
Army- AV, EA, MI, MR, TE
Navy- AS
Air Force- 03, 68

Preparing activity:
Army- AR
(Project: 6810-2018-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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